

Overall status: baseline models and prediction service are built; KERMT training, the front end and the final model comparison remain. MARS predicts 14 drug-property (ADMET) endpoints from a molecule structure. Data, the baseline model suite and the prediction service are built and tested. The advanced graph-neural-network (KERMT) track has begun training on GPU, and the user-facing web app and final evaluation have not started.

MILESTONE PROGRESS

Stage	Scope and current state	Status
M0	Contracts, scaffolding, CI, local stack	Complete
M1	Data acquisition, cleaning, splits and molecular featurization (14 datasets)	Complete
M2	Modelling: XGBoost baselines done; KERMT graph network partly trained	In progress
M3	Prediction API, accounts and storage, batch jobs, caching (built and tested locally); cloud deploy pending	Local complete
M4	Web front end, 3D molecule viewer, explainability	Not started
M5	Full evaluation and ablations, write-up, launch	Not started

MODEL TRAINING AND APPROXIMATE TIME PER SEED

Model	Coverage	Seeds	Approx. time per seed	Status
XGBoost baseline (laptop CPU)	All 14 endpoints, 70 runs, none failed	5 of 5	Smaller datasets ~15 s Medium ~35 s Large (CYP, hERG) ~1 min, up to ~6 min <i>Full sweep ~2.5 h of compute</i>	Complete
KERMT single-task: DILI liver injury (RTX A4000 GPU)	1 endpoint, first real GPU training, 22 Sep	5 of 5	~4 min per seed <i>~20 min for all five</i>	Trained
KERMT multi-task: hERG + AMES toxicity (RTX A4000 GPU)	2 endpoints trained jointly, 22 and 24 Sep	2 of 5	~95 min per seed <i>3 seeds remain, ~4.8 h (deferred)</i>	Partial
KERMT other 4 groups (absorption/distribution, metabolism)	Remaining endpoints	0	Not yet measured <i>est. ~2.5 to 3 h for one seed each</i>	Not started

COMPLETED

- **Data:** 14 public ADMET datasets acquired, standardised and split with leakage checks; molecular features cached.
- **Baseline models:** 5-seed XGBoost ensemble per endpoint with uncertainty and applicability-domain flags.
- **Held-out test results:** classification AUROC 0.86 to 0.97 across the 9 classification endpoints (e.g. CYP3A4 0.90, hERG 0.88, HIA 0.97); regression errors reported for the 5 regression endpoints.
- **Service:** prediction, compare and batch endpoints, sign-in and saved results, caching and rate limiting.
- **Quality:** 633 ML tests pass (5 known failures from a data-lockfile mismatch); CI configured; experiments tracked in Weights & Biases.
- **KERMT:** pretrained 70M-parameter model integrated; 7 GPU runs done. DILI test AUROC 0.86 (XGBoost 0.90), but on a smaller training set.

WORK REMAINING

- **Breadth-first KERMT training (new plan):** one seed of each of the 4 untrained groups first, then extra seeds up to 5 each.
- **Fair comparison:** align KERMT and XGBoost data set-up (KERMT DILI used a smaller training pool) and compare on the test set.
- **Calibration:** probability calibration on some endpoints is weaker than raw scores; the HIA calibration set is too small. Needs a decision.
- **Front end (M4):** web app, 3D viewer, explainability. Not started.
- **Deployment (M3):** cloud hosting and monitoring; needs credentials and approval.
- **Final evaluation (M5):** ablation study, significance tests, results tables, paper write-up and launch.

Schedule note. The plan targets 30 September. The modelling milestone ran past its dates because the mixed classification and regression groups needed a custom training approach, chosen deliberately in favour of correctness. The front end and final evaluation are the main risk to the date. Timings above are approximate, from run logs; estimates for untrained KERMT groups are extrapolated, not measured.